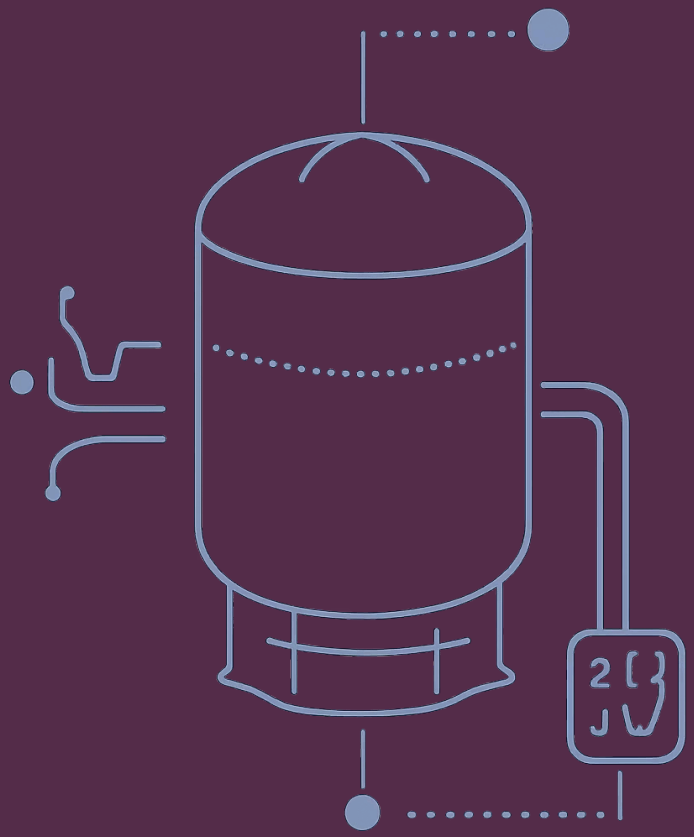
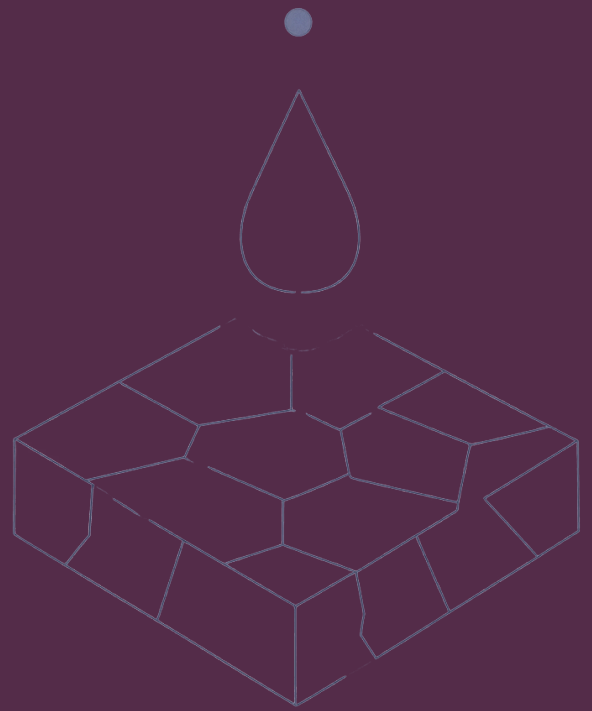


Smart Water Guardian

Automating Sustainable Resource Management through Combinational Logic

SUSTAINABLE ENERGY STARTUP

Made with GAMMA



The Crisis: A Silent Resource Drain

1% Fresh & Accessible

Of Earth's total water is fresh and accessible — a resource under severe pressure.

15–20% Wasted Daily

Manual tank management causes preventable overflow and dry-run losses every day.

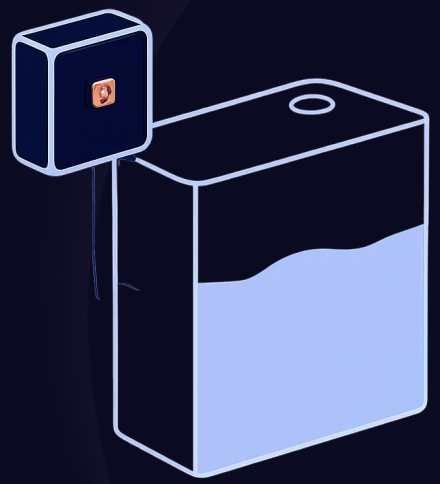
Hidden Energy Cost

Over-pumping depletes energy and causes costly infrastructure damage over time.

Made with GAMMA

Defining the Solution

SMART PROTECTION SYSTEM



Objective

Eliminate human intervention with precision automated control — no manual checks, no guesswork.

Core Logic

Sensors detect low and high-water thresholds, feeding digital signals directly into logic gates.

Key Benefit

Prevents dry-run pump damage and catastrophic overflow — protecting both equipment and water supply.

Made with GAMMA

Design Methodology: Combinational Logic

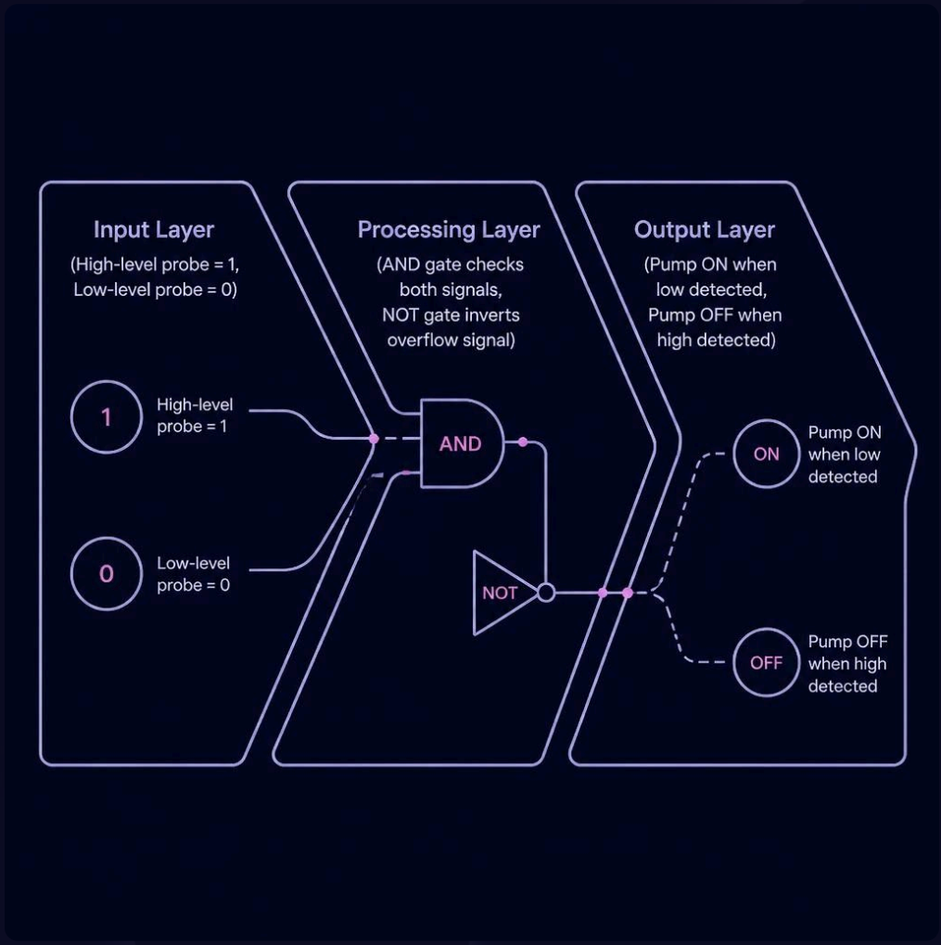
The system architecture is built on a foundation of pure digital logic — no microcontrollers, no software lag.



Sensor inputs feed directly into digital gates, enabling faster, more reliable, and lower-power decisions than processor-based alternatives.

Made with GAMMA

Circuit Logic: The Brain of the System



How It Decides

High-level and low-level probes provide binary digital signals. AND and NOT gates evaluate tank capacity in real time.

Output HIGH → Pump ON

Tank level is low; pump activates.

Output LOW → Pump OFF

High-level threshold reached; pump stops.

Made with GAMMA



Simulation Reveal: Testing in Logisim

Visual Current Flow

State transitions are demonstrated in real-time, making logic behavior visible and auditable.

Corner-Case Verification

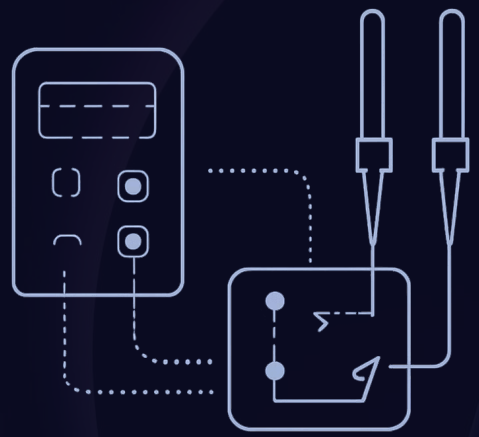
Tests include sudden sensor failure and simultaneous threshold triggers to validate robustness.

Pre-Build Validation

The full design is verified before any physical components are purchased or assembled.

Made with GAMMA

Hardware Implementation



Relay / Contactor

Acts as the physical switch for the 220V AC pump, controlled safely by low-voltage logic.

Conductivity Probes

Chosen for long-term stability and resistance to corrosion in wet environments.

Isolated Control Circuits

High-voltage and low-voltage sections are fully isolated to prevent electrical accidents.

Made with GAMMA

The Open-Source Ecosystem

GitHub Repository

Full schematics, Logisim logic files, and project documentation — publicly accessible and version-controlled.

Transparency

Enabling community review, modification, and improvement — building trust through openness.

Scalability

The base design can be adapted for multi-tank setups or industrial-scale water management.



Made with GAMMA

Strategic Impact for Sustainable Startups

Market Potential

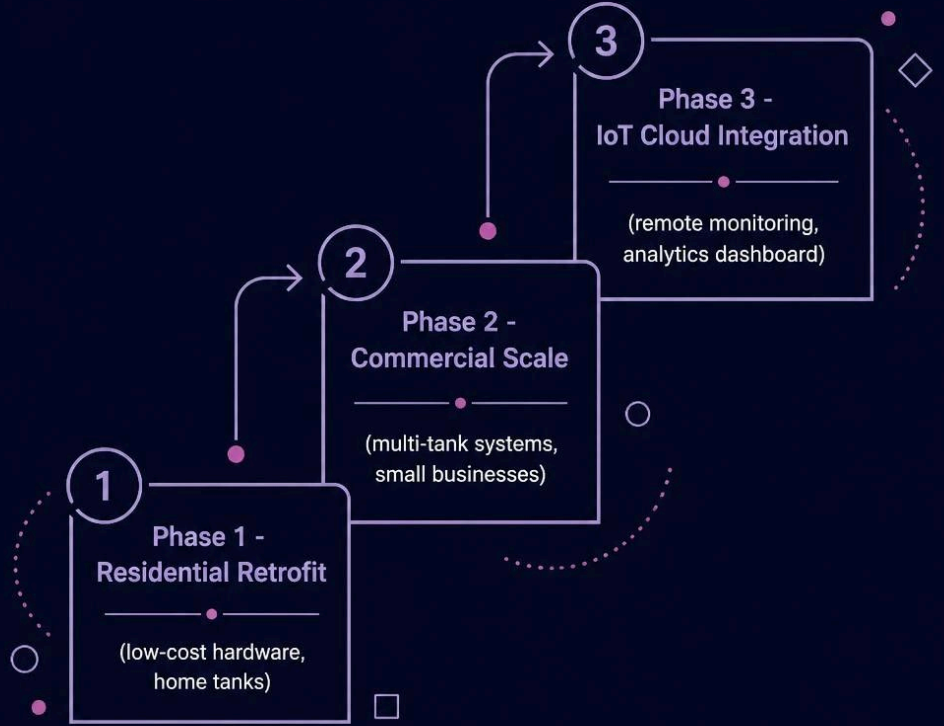
Retrofitting existing residential and commercial infrastructure with low-cost smart controls creates an immediate, addressable market.

Competitive Edge

Minimal hardware cost paired with significant resource savings delivers a compelling value proposition.

Long-Term Vision

IoT integration enables remote cloud monitoring, predictive maintenance, and data-driven water management at scale.



Made with GAMMA



Conclusion: A Future of Smart Conservation

🎯 The Goal

Move from manual dependence to intelligent, autonomous water management systems.

🤝 Call to Action

Join the open-source movement — explore, fork, and contribute via our GitHub repository.

💧 The Promise

Every drop saved is a measurable step toward global water security and sustainability.

GitHub Repository: Full schematics, Logisim files, and documentation available for the community.

Made with GAMMA